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gigs: A package for standardising fetal, neonatal, and child growth assessment

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Summary

There is a gap for clear guidance and open access tools for assessing nutrition and growth indicators as part of individual clinical care and population-based epidemiology. To enable better identification of at-risk infants, a unifying framework and clear guidance for how, when and for whom to use the existing international newborn and child growth standards developed by the WHO and INTERGROWTH-21st is needed. Such guidance is essential to enable tracking of progress towards Sustainable Development Goals and the WHO Global Nutrition targets. This need is the motivation for `gigs` - a package for researchers and policymakers that facilitates the appropriate use of growth standards for the determination and assessment of growth outcomes at the population level (e.g. stunting, wasting, underweight and trajectories over time) among vulnerable infants. For clinicians, the `gigs` also provides support in choosing which standards to use when assessing growth of individual infants.

Statement of need

Without accurate global data on newborn and child growth patterns using appropriate tools, true gaps and achievements in newborn and child health and survival will not be revealed. Those most vulnerable to poor growth outcomes will not be properly identified, tracking progress towards global targets (e.g. the Sustainable Development Goals) will be more difficult, and evaluating what changes are 'true progress' will be even more challenging. The `gigs` package allows researchers, policy-makers, and other key stakeholders to apply and correctly identify at-risk, vulnerable infants at a population level. This can then inform appropriate interventions to make progress towards global, national and local goals. Similarly, use of appropriate standards by clinicians will enable proper identification of at-risk infants and inform the clinical care management and decision-making.

Applying appropriate growth standards can be difficult - not all research groups have the ability to implement growth standards in software, and may make errors whilst doing so. Furthermore, researchers must pick and choose the right growth standards for their work - which is not always achieved in practice (Perumal et al., 2015). The `gigs` package makes these issues less common. Firstly, extensive unit testing ensures that each growth standard in `gigs` is accurate to published charts, for every growth standard we've implemented:

- The INTERGROWTH-21st Fetal Standards (Drukker et al., 2020; A. T. Papageorghiou et al., 2014; Aris T. Papageorghiou et al., 2014, 2016; Rodriguez-Sibaja et al., 2021; Stirnemann et al., 2017, 2020)
- INTERGROWTH-21st Newborn Size and Very Preterm Newborn Size Standards (Villar et al., 2014, 2016, 2017)

- INTERGROWTH-21st Postnatal Growth of Preterm Infants Standards (Villar et al., 2015)
- WHO Child Growth Standards (WHO Multicentre Growth Reference Study Group & de Onis, 2006; World Health Organisation, 2006, 2007)

This is more growth standards than made available in other R packages designed to perform the same sorts of analyses. For example, the WHO Child Growth Standards are implemented in many other packages: the `zanthro` package for Stata (Vidmar et al., 2013), and the R packages `anthro` (Schumacher, 2023), `AGD` (van Buuren, 2024), `childsds` (Vogel, 2022), `growthstandards` (Hafen, 2024), `sitar` (Cole, 2023), and `zscorer` (Myatt & Guevarra, 2019). However, only `gigs` implements the full set of INTERGROWTH-21st Fetal standards and INTERGROWTH-21st Postnatal Growth of Preterm Infants standards.

Next, `gigs` offers specific functions (`classify_growth()`, `gigs_waz()`, and friends) which select an appropriate growth standard to apply to each observation in a flat, tabular dataset. Growth standards are selected based on the chronological and gestational age for each observation, and the specific criteria can be found in the package documentation. These functions can be used to investigate growth indicators including size-for-gestational age, small vulnerable newborns (Ashorn et al., 2023; Lawn et al., 2023), stunting, wasting, weight-for-age (underweight), and micro/macrocephaly.

When benchmarked against each other, `gigs` for R outperforms almost all the packages mentioned above, or is at most a few milliseconds slower. It does this whilst checking inputs to ensure they are formatted correctly, to make life easier for `gigs` users. For example, `gigs` will detect missing (`NA`) or infinite (`-Inf`/`Inf`) numeric inputs, and flag these to the user in the console. Full benchmarking results are available on the package documentation website.

In addition to the R package, a Stata package and SAS package are available for users of those statistical applications. All versions of `gigs` are already in use by researchers at LSHTM, Harvard, the Small Vulnerable Newborn Collaboration and an international collaboration investigating stillbirths. Thanks to its speed, input checking, and unit test-backed accuracy, we believe `gigs` will be used widely by those looking to assess newborn, infant, or child growth in R.

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References

Ashorn, P., Ashorn, U., Muthiani, Y., Aboubaker, S., Askari, S., Bahl, R., Black, R. E., Dalmiya, N., Duggan, C. P., Hofmeyr, G. J., Kennedy, S. H., Klein, N., Lawn, J. E., Shiffman, J., Simon, J., Temmerman, M., Okwaraji, Y., Krasevec, J., Bradley, E., … Hayashi, C. (2023). Small vulnerable newborns—big potential for impact. *The Lancet*, *401*(10389), 1692–1706. https://doi.org/10.1016/S0140-6736(23)00354-9

Cole, T. (2023). *Sitar: Super imposition by translation and rotation growth curve analysis*. https://doi.org/10.32614/CRAN.package.sitar

Drukker, L., Staines-Urias, E., Villar, J., Barros, F. C., Carvalho, M., Munim, S., McGready, R., Nosten, F., Berkley, J. A., Norris, S. A., Uauy, R., Kennedy, S. H., & Papageorghiou, A. T. (2020). International gestational age-specific centiles for umbilical artery Doppler indices:

A longitudinal prospective cohort study of the INTERGROWTH-21st Project. *American Journal of Obstetrics & Gynecology*, *222*(6), 602.e1–602.e15. https://doi.org/10.1016/j.ajog.2020.01.012

Hafen, R. (2024). *Growthstandards: Anthropometric growth standard calculations*. https://github.com/ki-tools/growthstandards

Lawn, J. E., Ohuma, E. O., Bradley, E., Idueta, L. S., Hazel, E., Okwaraji, Y. B., Erchick, D. J., Yargawa, J., Katz, J., Lee, A. C. C., Diaz, M., Salasibew, M., Requejo, J., Hayashi, C., Moller, A.-B., Borghi, E., Black, R. E., Blencowe, H., Ashorn, P., … Prendergast, A. (2023). Small babies, big risks: global estimates of prevalence and mortality for vulnerable newborns to accelerate change and improve counting. *The Lancet*, *401*(10389), 1707–1719. https://doi.org/10.1016/S0140-6736(23)00522-6

Myatt, M., & Guevarra, E. (2019). *Zscorer: Child anthropometry z-score calculator*. https://doi.org/10.32614/CRAN.package.zscorer

Papageorghiou, A. T., Kennedy, S. H., Salomon, L. J., Ohuma, E. O., Cheikh Ismail, L., Barros, F. C., Lambert, A., Carvalho, M., Jaffer, Y. A., Bertino, E., Gravett, M. G., Altman, D. G., Purwar, M., Noble, J. A., Pang, R., Victora, C. G., Bhutta, Z. A., Villar, J., & International Fetal and Newborn Growth Consortium for the 21st Century (INTERGROWTH-21st). (2014). International standards for early fetal size and pregnancy dating based on ultrasound measurement of crown–rump length in the first trimester of pregnancy. *Ultrasound in Obstetrics & Gynecology*, *44*(6), 641–648. https://doi.org/10.1002/uog.13448

Papageorghiou, Aris T., Ohuma, E. O., Altman, D. G., Todros, T., Cheikh Ismail, L., Lambert, A., Jaffer, Y. A., Bertino, E., Gravett, M. G., Purwar, M., Noble, J. A., Pang, R., Victora, C. G., Barros, F. C., Carvalho, M., Salomon, L. J., Bhutta, Z. A., Kennedy, S. H., & Villar, J. (2014). International standards for fetal growth based on serial ultrasound measurements: The Fetal Growth Longitudinal Study of the INTERGROWTH-21st Project. *The Lancet*, *384*(9946), 869–879. https://doi.org/10.1016/S0140-6736(14)61490-2

Papageorghiou, Aris T., Ohuma, E. O., Gravett, M. G., Hirst, J., Silveira, M. F. da, Lambert, A., Carvalho, M., Jaffer, Y. A., Altman, D. G., Noble, J. A., Bertino, E., Purwar, M., Pang, R., Cheikh Ismail, L., Victora, C., Bhutta, Z. A., Kennedy, S. H., & Villar, J. (2016). International standards for symphysis-fundal height based on serial measurements from the Fetal Growth Longitudinal Study of the INTERGROWTH-21st Project: Prospective cohort study in eight countries. *BMJ*, *355*, i5662. https://doi.org/10.1136/bmj.i5662

Perumal, N., Gaffey, M. F., Bassani, D. G., & Roth, D. E. (2015). WHO Child Growth Standards Are Often Incorrectly Applied to Children Born Preterm in Epidemiologic Research. *The Journal of Nutrition*, *145*(11), 2429–2439. https://doi.org/10.3945/jn.115.214064

Rodriguez-Sibaja, M. J., Villar, J., Ohuma, E. O., Napolitano, R., Heyl, S., Carvalho, M., Jaffer, Y. A., Noble, J. A., Oberto, M., Purwar, M., Pang, R., Cheikh Ismail, L., Lambert, A., Gravett, M. G., Salomon, L. J., Drukker, L., Barros, F. C., Kennedy, S. H., Bhutta, Z. A., & Papageorghiou, A. T. (2021). Fetal cerebellar growth and Sylvian fissure maturation: International standards from Fetal Growth Longitudinal Study of INTERGROWTH-21st Project. *Ultrasound in Obstetrics & Gynecology*, *57*(4), 614–623. https://doi.org/10.1002/uog.22017

Schumacher, D. (2023). *Anthro: Computation of the WHO child growth standards*. https://doi.org/10.32614/CRAN.package.anthro

Stirnemann, J., Salomon, L. J., & Papageorghiou, A. T. (2020). INTERGROWTH-21st standards for Hadlock's estimation of fetal weight. *Ultrasound in Obstetrics & Gynecology*, *56*(6), 946–948. https://doi.org/10.1002/uog.22000

Stirnemann, J., Villar, J., Salomon, L. J., Ohuma, E., Ruyan, P., Altman, D. G., Nosten, F., Craik, R., Munim, S., Cheikh Ismail, L., Barros, F. C., Lambert, A., Norris, S.,

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Carvalho, M., Jaffer, Y. A., Noble, J. A., Bertino, E., Gravett, M. G., Purwar, M., … Papageorghiou, A. T. (2017). International estimated fetal weight standards of the INTERGROWTH-21st Project. *Ultrasound in Obstetrics & Gynecology*, *49*(4), 478–486. https://doi.org/10.1002/uog.17347

van Buuren, S. (2024). *AGD: Analysis of growth data*. https://doi.org/10.32614/CRAN.package.AGD

Vidmar, S. I., Cole, T. J., & Pan, H. (2013). Standardizing anthropometric measures in children and adolescents with functions for egen: Update. *Stata Journal*, *13*(2), 366–378. https://doi.org/10.1177/1536867X1301300211

Villar, J., Giuliani, F., Bhutta, Z. A., Bertino, E., Ohuma, E. O., Ismail, L. C., Barros, F. C., Altman, D. G., Victora, C., Noble, J. A., Gravett, M. G., Purwar, M., Pang, R., Lambert, A., Papageorghiou, A. T., Ochieng, R., Jaffer, Y. A., Kennedy, S. H., Katz, M., … Enquobahrie, D. (2015). Postnatal growth standards for preterm infants: The Preterm Postnatal Follow-up Study of the INTERGROWTH-21st Project. *The Lancet Global Health*, *3*(11), e681–e691. https://doi.org/10.1016/S2214-109X(15)00163-1

Villar, J., Giuliani, F., Fenton, T. R., Ohuma, E. O., Ismail, L. C., & Kennedy, S. H. (2016). INTERGROWTH-21st very preterm size at birth reference charts. *The Lancet*, *387*(10021), 844–845. https://doi.org/10.1016/S0140-6736(16)00384-6

Villar, J., Ismail, L. C., Victora, C. G., Ohuma, E. O., Bertino, E., Altman, D. G., Lambert, A., Papageorghiou, A. T., Carvalho, M., Jaffer, Y. A., Gravett, M. G., Purwar, M., Frederick, I. O., Noble, A. J., Pang, R., Barros, F. C., Chumlea, C., Bhutta, Z. A., Kennedy, S. H., … Enquobahrie, D. (2014). International standards for newborn weight, length, and head circumference by gestational age and sex: The Newborn Cross-Sectional Study of the INTERGROWTH-21st Project. *The Lancet*, *384*(9946), 857–868. https://doi.org/10.1016/s0140-6736(14)60932-6

Villar, J., Puglia, F. A., Fenton, T. R., Cheikh Ismail, L., Staines-Urias, E., Giuliani, F., Ohuma, E. O., Victora, C. G., Sullivan, P., Barros, F. C., Lambert, A., Papageorghiou, A. T., Ochieng, R., Jaffer, Y. A., Altman, D. G., Noble, A. J., Gravett, M. G., Purwar, M., Pang, R., … Bhutta, Z. A. (2017). Body composition at birth and its relationship with neonatal anthropometric ratios: the newborn body composition study of the INTERGROWTH-21st project. *Pediatric Research 2017 82:2*, *82*(2), 305–316. https://doi.org/10.1038/pr.2017.52

Vogel, M. (2022). *Childsds: Data and methods around reference values in pediatrics*. https://doi.org/10.32614/CRAN.package.childsds

WHO Multicentre Growth Reference Study Group, & de Onis, M. (2006). WHO Child Growth Standards based on length/height, weight and age. *Acta Paediatrica Supplement*, *450*, 76–85. https://doi.org/10.1111/j.1651-2227.2006.tb02378.x

World Health Organisation. (2006). *WHO child growth standards: length/height-for-age, weight-for-age, weight-for-length, weight-for-height and body mass index-for-age: methods and development* (pp. 1–312). World Health Organisation. ISBN: 924154693X

World Health Organisation. (2007). *WHO child growth standards: head circumference-for-age, arm circumference-for-age, triceps skinfold-for-age and subscapular skinfold-for-age: methods and development* (pp. 1–237). World Health Organisation. ISBN: 9789241547185

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